

File Name	Brief Description
app.py	Main flask application file. Contains the web routes, user registration/login logic, session handling, flight search, ticket purchase logic, customer/agent/staff dashboards, staff admin tools, and database query logic.
db.py	Required database connection helper. Defines get_connection(), which connections the Flask app to the MySQL database using PyMySQL and the settings from config.py
config.py	Centralized configuration file. Stores the Flask secret key and database connection settings, including host, port, username, password, and database name.
requirements.txt	Lists the Python packages required to run the project, including Flask, PyMySQL, and Werkzeug.
Templates Folder	HTML Files
base.html	Main shared layout used by the site. Defines the page structure, navigation bar, dropdown registration links, login/logout display, session-based dashboard links, flash message display, and the Jinja content block used by all child pages.
index.html	Public home page for the Air Ticket Reservation System. Provides basic navigation links for searching flights, checking flight status, registering as a customer/agent/staff member, and signing in.
login.html	Public login page used by customers, agents, and airline staff.
register_staff.html	Registration page for airline staff. Includes the option to register as either Admin or Operator.
customer_dashboard.html	Landing page for logged-in customers. Includes links to view purchased flights, public flight search, and spending review.
agent_dashboard.html	Landing page for logged-in booking agents. Includes links to view flights booked by the agent, public flight search, and commission analytics.
staff_dashboard.html	Landing page for logged-in airline staff. Includes list of all flights operated by the airline they work for in the next 30 days, passenger lists for flights, various statistics, and other functions depending on the staff's permissions (Admin or Operator).
register_agent.html	Registration page for booking agents.
register_customer.html	Registration page for customers.
search_flights.html	Public flight search page, can filter upcoming flights by origin or destination airport, city, and date.
flight_status.html	Public page to search an upcoming flight and see its status.
customer_flights.html	Customer-specific page to see upcoming purchased flights, can filter by date ranges, origin, and destination.

customer_spending.html	Customer-specific page to view spending on flights. Can filter by a date range.
agent_analytics.html	Agent-specific page to access their analytics such as commission totals for last 30 days, average commission per ticket, number of tickets sold, bar charts showing top 5 customers by different metrics.
staff_analytics.html	Staff-specific page to show airline analytics.
staff_passengers.html	Staff-specific page to show passenger list per flight operated by the airline the staff works for.
staff_customer_flights.html	Staff-specific page to show all flights taken by a specific customer on the airline the staff works for. Cannot access passengers that don't exist or passengers that have not taken a flight with the airline.
staff_agents_table_simple.html	Reusable partial template used by staff_analytics.html to display booking agent ranking tables, including agent email, tickets sold, and commission. Shows a fallback message when no agent data is available.
staff_add_airport.html	Admin-specific page to add an airport. Includes fields to input airport city name and three-letter airport code.
staff_add_airplane.html	Admin-specific page to add an airplane.
staff_create_flight.html	Admin-specific page to create a flight for the airline they work for.
staff_authorize_agent.html	Admin-specific page to authorize a booking agent for the airline.
staff_update_status.html	Operator-specific page to update the status of flights.
staff_add_airport_result.html	Success page shown when a new airport is successfully added.
agent_flights.html	Booking agent-specific page to show all flights the agent has booked for customers.
agent_book_flight.html	Booking agent-specific page that displays the selected flight's details, lets the agent book a flight for a customer, and provides a dropdown of available seat classes with calculated prices and remaining seat counts.